04. EYE PLACEMENT

DURATION : 08:39 STUDY SHEET

COURSE SUMMARY

This video explores the development of the eye within the framework of biodynamic embryology and osteopathy. You will learn how the interaction between the endoderm and mesoderm, as well as the primary role of the notochord and neural tube, contribute to the formation of essential structures such as the eye. The movements of permeation and infusion, along with the concept of polarity, are key concepts underlying embryonic development. Furthermore, it will be discussed how rebalancing techniques, in response to emotional or physical whiplash, can restore the movements necessary for optimal eye development, thus linking it to events such as the first heartbeat.

EYE DEVELOPMENT

The **endoderm** is an epithelium, while the **mesoderm** is a connective tissue. It is essential to understand the balance between these two major tissue types: an internal tissue and an external tissue. For example, the digestive epithelium, although internal, acts as a boundary tissue.

The precursor to the **neural groove** and the **neural plate** is the **notochord**. This process of evagination along a longitudinal axis is related to the **primitive streak**, which forms the primitive axis of the body. This axis is crucial because it organises cells according to their space, time, and destiny.

By day 18, the neural plate develops, and the third cavity, that of the **external coelom**, appears. This change in polarity and the reorganisation of the axis create the primitive streak through a suction field from the caudal region, which "aspirates" information. This phenomenon is accompanied by movements called **permeation** and **infusion**.

The permeation movement, a major metabolic movement, occurs around the embryo, advancing the **amniotic cavity**. Thus, we distinguish the amniotic cavity, the vitelline cavity, and a tissue called the **epiblast**, which will give rise to the future nervous system, while the tissue beneath it will form the future digestive system.

This movement causes a change in the shape of the notochord, which takes on an S-shape. This differential growth is linked to polarity. The primitive axis organises the epithelial tissue, which reacts through phenomena of induction and inhibition in relation to the notochord.

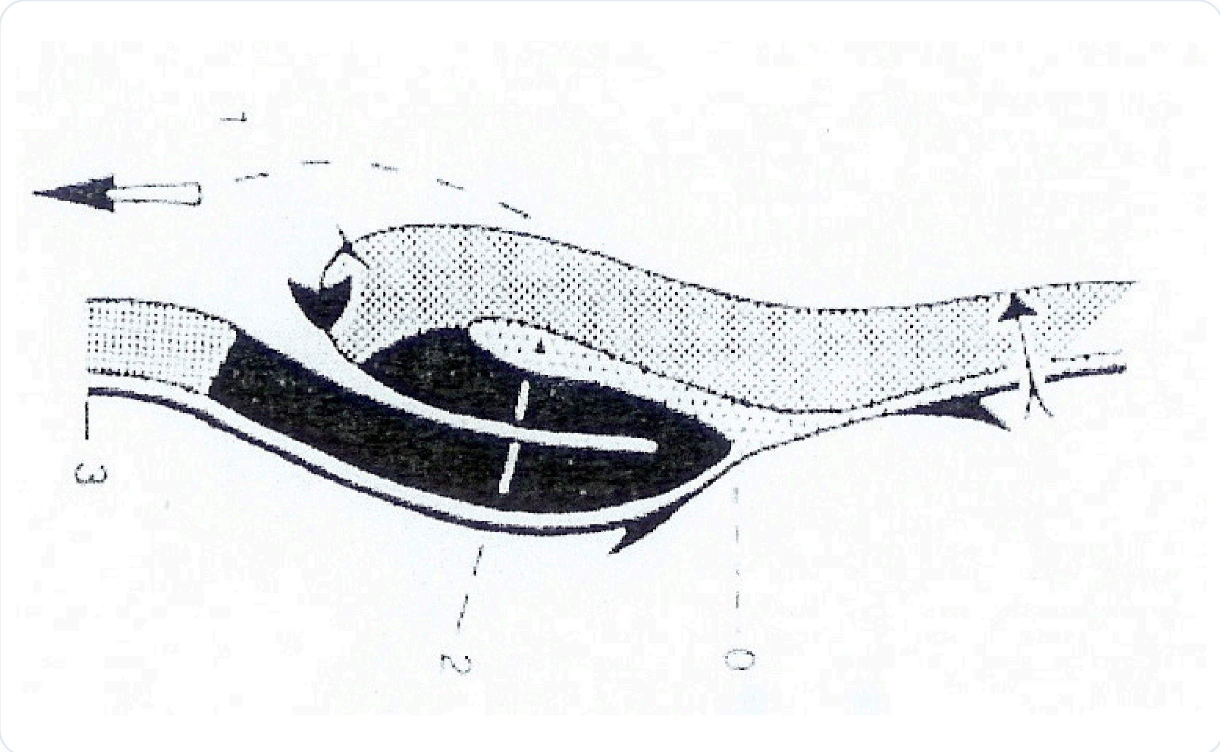

FIG. — Differential growth

The neural plate evolves to become a **neural groove**, which encloses cerebrospinal fluid, or rather primitive amniotic fluid. The closure of this groove forms a **neural tube**. The **neural crests**, considered a fourth embryonic tissue, play a crucial role in eye development.

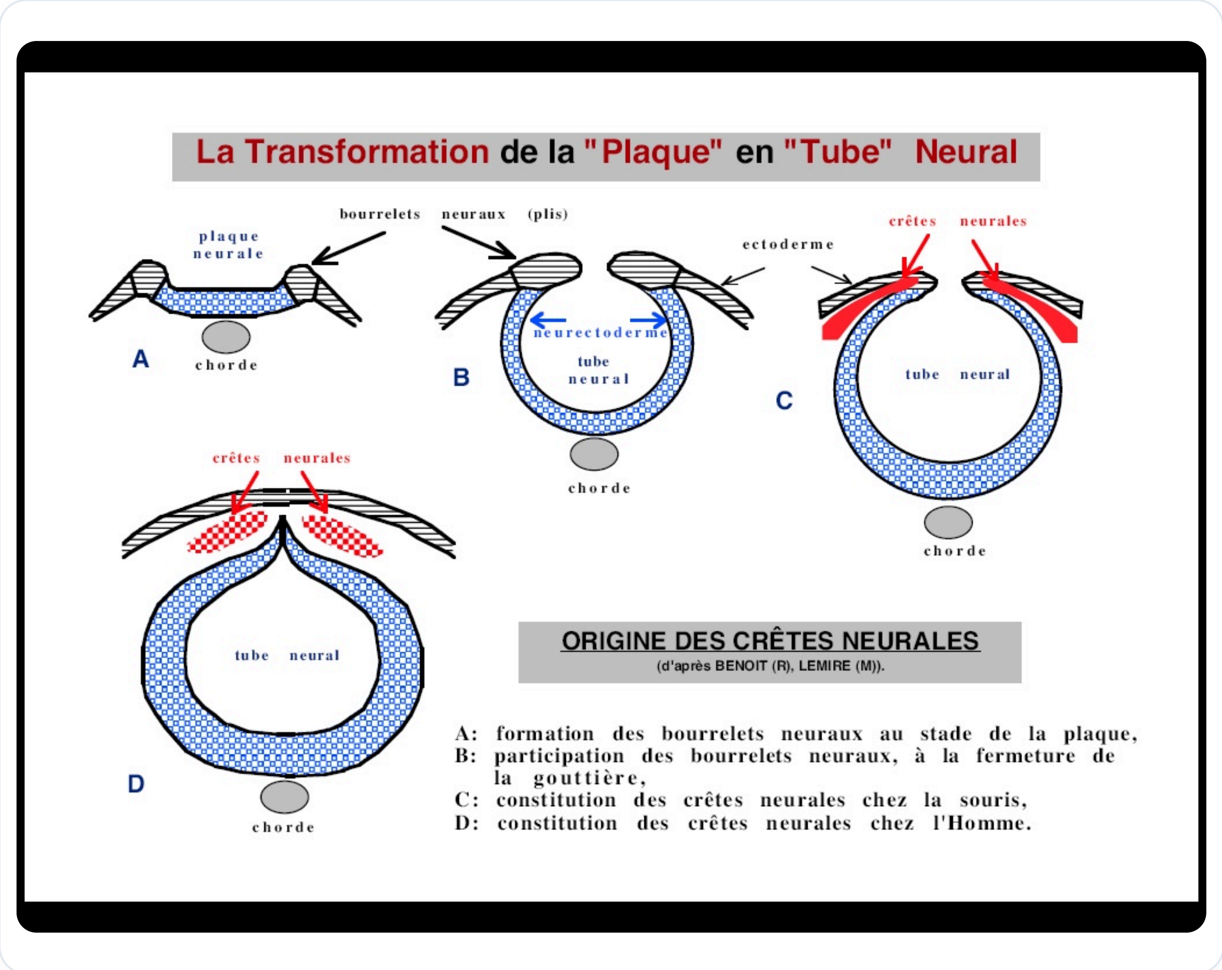

FIG. — Primitive amniotic fluid

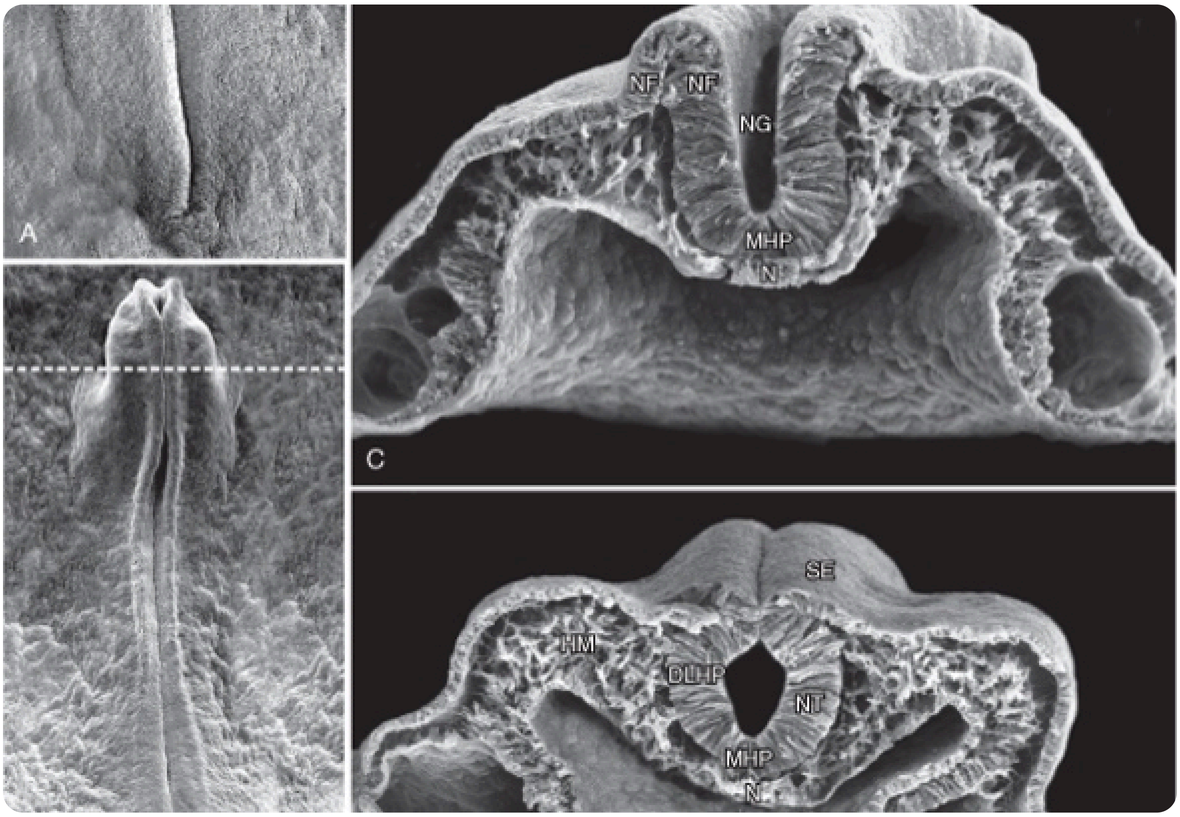

FIG. — Neural groove formation

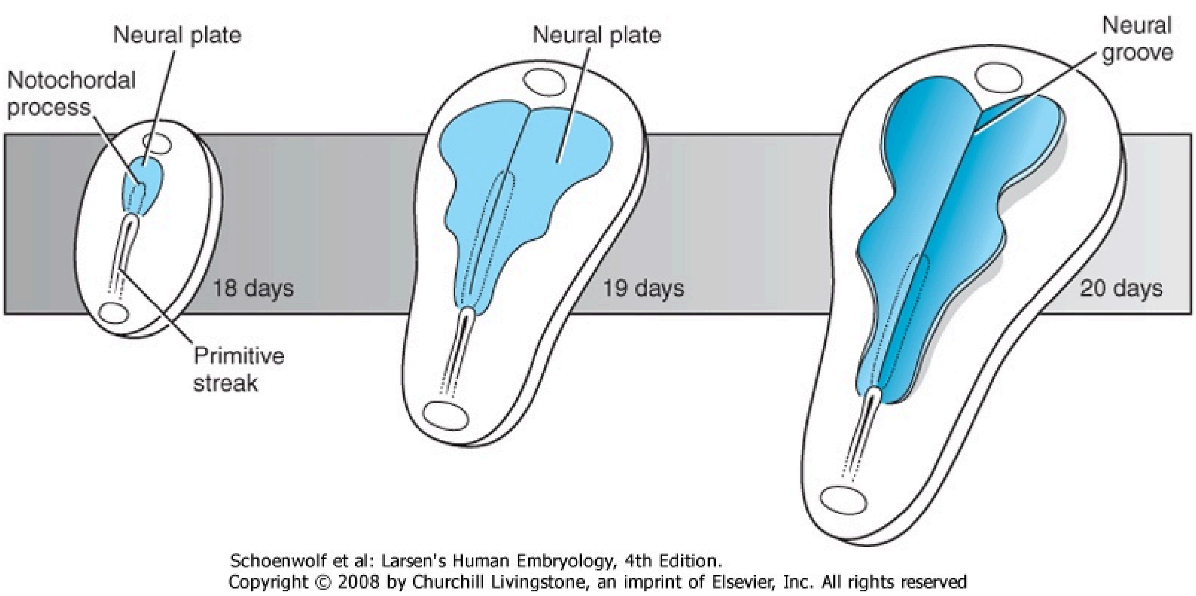

FIG. — Transformation of the neural plate into the neural groove

The notochord is essential for eye development. Without a balanced notochord, it is impossible to have a stable eye. Similarly, a free sacrum is necessary to ensure brain stability. It is therefore fundamental to rebalance the sacrum to ensure good verticality.

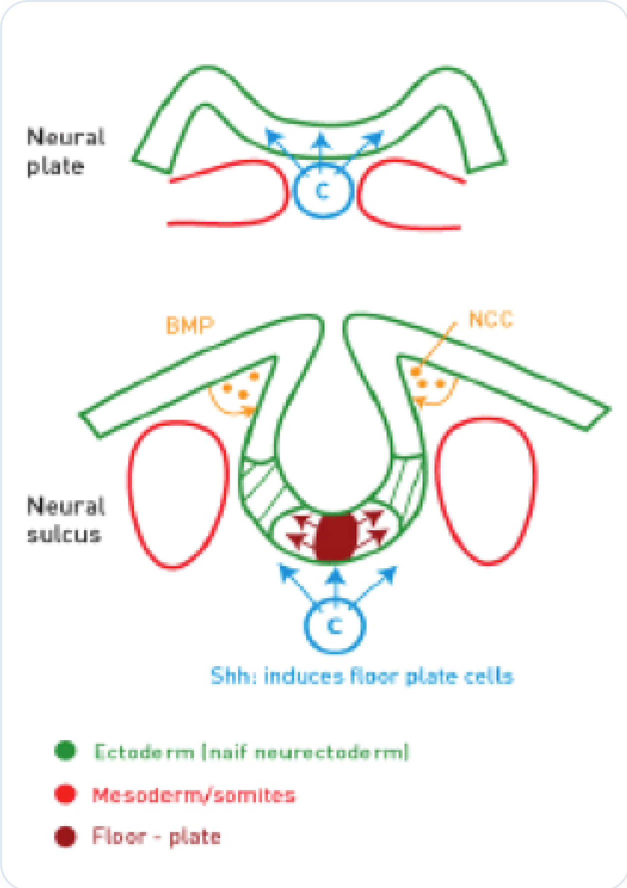

FIG. — Closure to form the neural tube and neural crest formation

The embryo presents the amniotic cavity, the vitelline cavity, and the bilaminar disc, which organise around the notochordal axis and the neural tube. The formation of this neural tube involves the closure of the amniotic cavity and the vitelline cavity.

As the longitudinal axis of the embryo develops, the brain anlage forms and closes at the centre of the embryo, leaving a remnant at the top. It is in this space that the **lens placode** develops, which will form the eye.

Before the closure of the neural tube, an induction phenomenon occurs to form the notochord, which is the basis of cortical development. From this development will emerge the eye, which is an expansion of a ventricular system filled with cerebrospinal fluid. The eye can be considered a fluid sphere, a subtle filter for photons, and a balancer of the surrounding space.

It is possible to rebalance the eye by working on a specific area, often affected by **whiplash**. These whiplash injuries can be emotional or physical. Treating a whiplash involves restoring the movements of **cephalisation**, **cardialisation**, **diaphragmatisation**, and **hepatisation**, which are essential for embryonic development.

The emergence of the eye begins with the first heartbeat, around day 22. The first epiblastic cell transformation occurs in synchronisation with these primitive heartbeats, establishing a significant correspondence between eye development and cardiac activity.